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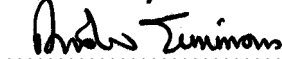
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METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH TI 21 TRACK CIRCUITS

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The provisions of this code of practice may be used from the date of issue.

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I. INTRODUCTION

I.1 Purpose

The purpose of this procedure is to provide a method for calculating the safe permissible maximum levels of electrical interference that may be generated by electric trains designed to be used on tracks employing TI 2I track circuits. The procedure lists all of the infrastructure aspects to be taken into account, characteristics of the TI 2I track circuits and possible failure modes. Worked examples are given for calculating maximum permissible levels of electrical interference due to traction current, and axle-to-axle voltages.

I.2 Scope

The scope of this procedure is that of providing a method for presenting calculations in support of a safety case for the operation of electric trains over TI 2I track circuits. All components relevant to TI 2I track circuits are considered.

The document only considers double rail configuration with balanced earth connections.

The electrification installed is assumed to be of the standard arrangement, auto-transformer fed sections of railway are not included.

Traction bonding is assumed to be compliant with the requirements of RT/E/PS/11761 [2].

I.3 Installation types covered in this document

This procedure may be used for the operation of trains over Railtrack 25 kV a.c. or 750 V d.c. electrified routes. The levels derived define the maximum permitted interference generated by a single train. Double rail end-fed and centre fed TI 2I track circuits are considered.

On a.c. electrified routes the catenary support and other lineside metal structures are bonded to traction return rails. This document relates to double rail installations only and these are nominally balanced. The TI 2I track circuit requires both rails to be balanced with respect to voltage to earth and current flow in each rail. Therefore, in TI 2I areas the catenary support structures are often bonded to a dedicated earth wire connected to the rail via impedance bonds [1].

On d.c. electrified routes neither of the rails is intentionally bonded to earth, hence balance is maintained. In a completely balanced situation, the TI 2I track circuit is in theory immune to any amount of traction current or interference produced by a train.

1.4 Equipment types included in this document

The equipment types included in this document are the type designated TI2I-1 used in the UK. These are assumed to be installed only in double rail mode in jointless configuration.

2 SOURCE DOCUMENTS

2.1 Relevant Railway Standards

1. 'Track Circuit Requirements for AC Traction - Code of Practice', Safety & Standards Directorate, Railtrack, Fitzroy House, 355 Euston Road, London NW1 3AG, Reference GK/RC 0753, Issue 1, June 1996.
2. 'Railtrack line Product Specification TI.2I Audio Frequency Track Circuits.', Railtrack, Reference RT/E/PS/11761, Issue 1, December 2000.
3. 'Electrified Lines Traction Bonding', Safety & Standards Directorate, Railtrack, Fitzroy House, 355 Euston Road, London NW1 3AG, Reference GL/RT 1254, Issue 1, April 2000

2.2 Previous Safety case documentation

4. 'GEC Alsthom Juniper Project TI2I Track Circuit Interference Limits AC and DC Routes' AEA Technology Rail, Issue 2, 7 January 1999, Report reference PRJ-SCSE(JUN)-RP-7
5. 'Track parameters for Interference studies' N Rambukwella, AEAT Rail Report number STG-D-EPS(373)RP-18, Issue 1, January 1995.
6. 'Class 92 Locomotive EMC Safety Assessment – Volume 5 TI2I Track Circuits, Fleet Operation on the Route between Crewe and Doncaster', AEA Technology ref. 120-dr0459, version b0, dated February 1999.

7. "TI2I Track Circuit Proof of Safety", ABB Signal Ltd, dated 11th October 1993.

3. INFRASTRUCTURE ASSUMPTIONS

The main assumptions that may be made while carrying out an analysis are summarized below.

- For a track circuit with a TTU (Track Tuning Unit), the analyses are carried out for each of the TI 2I frequencies, abutting the other frequency of the allowable pairing at a tuning unit.
- For a track circuit with an ETU (End Termination Unit), the analyses are carried out for each of the TI 2I frequencies.
- A rail resistance of $1 \Omega/\text{km}$ and rail inductance of $675 \mu\text{H}/\text{km}$ is used over the TI 2I frequency band. This is in accordance with measured parameters for BS113a rail with an allowance for jointed track [4, 5].
- 100% imbalance is assumed, this is pessimistic compared to the nominal imbalance values of 60%/40% in the case of 25 kV a.c. power supplies and 70%/30% in the case of 750 V d.c. power supplies. [4]
- Other track circuits may be assumed to be balanced. Interference current from trains on other track circuits will therefore be significantly less than that on the track being analysed, and will be covered by an interference allowance.
- The impedance of the cables used for cross bonding is ignored.
- The reduction of rail current due to earth return currents (either intentional or due to leakage) can be ignored. This means that all return current is in the rails. Worst-case interference propagation is therefore considered.
- The effect of multiple interfering sources may be ignored since these are covered by interference allowances.
- All track circuit components are assumed to be installed to Railway Group standards and components are at their nominal values.
- Bond types ABB B3/500 and Westinghouse Style S are analysed for a.c. and d.c. traction systems respectively. These are considered to be representative.
- The maximum length considered is that stated in ref. 2 that is 1 km.

- The receiver is assumed to be set at its most sensitive setting (there are 13 settings in total).
- The sleepers in use are assumed to be concrete or wood, steel sleepers are not included in the analysis presented here.
- UIC 60 Rail is not included in the assessment as presented. The methodology given here is applicable but the parameters of this rail type at the frequency of interest will need to be determined and used in the appropriate calculation.
- Track boosting capacitances are excluded from the scope of this assessment.
- Traction bonding is assumed to be compliant with the requirements of RT/E/PS/11761 [2].

The following assumptions are made in the assessment of axle-to-axle voltages:

- The train producing axle-to-axle voltage does so at a spacing equal to the cross-bond spacing. Current flow in the train and track under the train are not considered. [4]
- Minimum cross-bond spacing is 100 m. [4]
- The train producing interference is in a track adjacent to that on which the track circuit in question is installed. This being the only parallel track.

3.1 Imbalance

The double rail TI 21 track circuit is intended to operate as a balanced system. For all current 25 kV schemes using this type of track circuit, the balance is maintained by avoiding any connections to the rails that could unbalance the track circuit. This is achieved by using an earth wire for structure bond connections. Earthing and traction return connections to the running rails are via impedance bonds. However, due to the different mutual inductances between each of the traction return rails and the earth wire, an imbalance of traction return current (and hence train interference current) will occur. The probable value of this imbalance is noted in the assumptions above. It is further assumed that the imbalance component of the noise current will flow in the TI 21 track circuits.

In third rail d.c. electrified areas, none of the rails are earthed and the traction return connections are made via TI 21 impedance bonds. When the track circuit is fully balanced, any traction return current will be presented equally to each input terminal of the track circuit receiver, and will have no effect on its operation. A wrong side

track circuit failure due to interference can only occur when there is some imbalance present.

Imbalance at track circuit frequencies can be caused by:

- 1 Unequal earthing of each rail through the ballast
- 2 Presence of check rails or similar
- 3 Differences in electrical characteristics of rails, bonding differences between each rail or series bonding
- 4 Earthing of either rail
- 5 Asymmetrical position of the conductor rail or catenary relative to the running rails
- 6 Track curvature

A broken rail (i.e. an electrical gap in continuity) will probably cause a right side failure of the track circuit. This will therefore be detected in a short time. The other causes of imbalance will be less significant, but the actual amount of imbalance is difficult to predict.

The presence of cross bonding has no direct effect on the track circuit, since all cross bonds are connected via impedance bonds. Whilst the impedance between the cross-bond center tap on the impedance bond and either rail, being low at traction current frequencies (d.c. or 50 Hz), it is somewhat higher at TI 2I frequencies. If a short circuit develops within the impedance bond, which connects the cross bond directly to one rail, the resonant frequency of the impedance bond will be reduced. The impedance at TI 2I frequencies will reduce significantly, and the track circuit will be shunted. This will cause a right side failure and it will be detected.

At points and crossings the ends of a TI 2I section will be connected to other types of track circuits via impedance bonds, therefore any failure of these bonds will also cause a right side failure. In a balanced system, there is little possibility of interference currents circulating via the cross bonding. This is because of the relatively high impedance of the impedance bonds at TI 2I frequencies. Since an impedance bond failure will be detected, multiple failures are considered incredible and are therefore not analysed. Rail impedance at TI 2I frequencies is relatively high, and any unbalanced interference generated by trains outside the track circuit being analysed will be attenuated by the rail impedance and shunted by the tuning units.

Thus, only one failure mode needs to be analysed: a train occupying a track circuit with 100% imbalance.

At TI 2I frequencies (nominally 2 kHz), a rail resistance of $1 \Omega/\text{km}$ and a total rail inductance of $675 \mu\text{H}/\text{km}$ are used in the calculations. A rail resistance of $0.72 \Omega/\text{km}$

and a rail inductance of 677 $\mu\text{H}/\text{km}$ [4], have also been obtained from a series of tests over a range of frequencies.

The 1 Ω/km rail resistance is considered in this procedure to be a pessimistic assumption to take into account the effect of rail bonds in a jointed track.

The effects of ballast resistance are negligible, as it is assumed to be infinite, although the effects of ballast capacitance are considered in the analysis. An infinite ballast resistance is the worst case for interference propagation in this analysis. The presence of ballast resistance will shunt the interference current and could reduce the amount of interference reaching the receiver [4].

4. DESCRIPTION OF TRACK CIRCUIT

The TI 21 is a jointless track circuit operating in the audio frequency band, in the range 1532 Hz to 2610 Hz. TI 21 track circuit equipment fitted to Railtrack controlled infrastructure is given the designation TI 21-I to differentiate it from TI 21 equipment that is used in other countries. The primary differences between the different types are operating frequencies and modulation rates. The equipment considered comprises the following:

- TI 21-I Transmitter
- TI 21-I Receiver & relay
- Track tuning unit
- Impedance bond
- End termination unit

4.1 Track Circuit Frequency Range and Bandwidth

Each track circuit uses two frequencies separated by 34 Hz, as shown below in Table I. These are chosen to provide isolation between adjacent track circuits.

The bandwidth is:

- -3 dB at ± 6 Hz from each centre frequency
- -20 dB at ± 30 Hz from each centre frequency

Track Circuit Frequency Code	Nominal Frequency, Hz	Operating Frequency, Hz	
		Lower Frequency	Upper Frequency
A	1699	1682	1716
B	2296	2279	2313
C	1996	1979	2013
D	2593	2576	2610
E	1549	1532	1566
F	2146	2129	2163
G	1848	1831	1865
H	2445	2428	2462

Table 1: TI 21 Track circuit operating frequencies

4.1.1 Frequency Allocation

Two nominal frequencies are used to signal each track, these are alternated to provide isolation. The permissible frequency combinations and allocations are shown in table 2, reproduced from [2].

Track Circuit Frequency Code	Use	
A	Down line	Frequencies for use in two track areas
B	Down line	
C	Up line	
D	Up line	
E	Down line	Additional frequencies for use in four track areas
F	Down line	
G	Up line	
H	Up line	

Table 2: TI 21 Track circuit channel allocations

4.2 Track circuit components

4.2.1 TI 2I Transmitter

Although the transmitter (Tx) is not considered in the interference assessment, details are included here for clarity in describing operation. A block diagram of the TI 2I transmitter is shown in Figure 1.

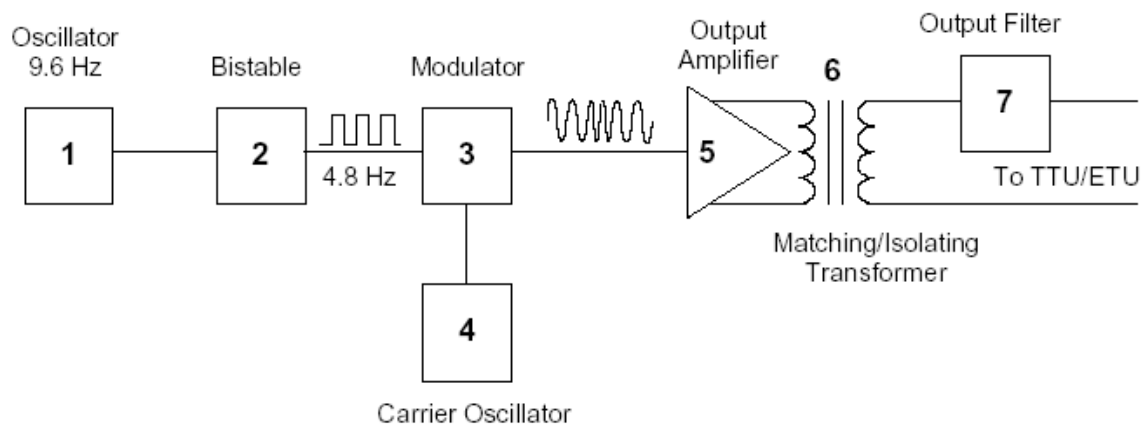


Figure 1: Block diagram of the TI 2I transmitter

The bistable (2) converts the output of the relaxation oscillator (1) from 9.6 Hz into a 4.8 Hz square wave. This frequency is used by the modulator (3) to modulate the output of the oscillator (4). The oscillator produces a signal that alternates ± 17 Hz either side of a carrier frequency at a rate of 4.8 Hz. The amplifier (5) increases this signal to a power level suitable for transmission to the track. A transformer (6) matches the amplifier output to the load. The filter (7) isolates the unit from unwanted d.c. and a.c. voltages. The signal is fed via a two-core cable to the associated Track Tuning Unit or End Termination Unit (TTU/ETU).

4.2.2 TI 2I Receiver and Relay

The track frequency is fed via a TTU or ETU to the receiver (Rx) which drives an a.c. immune, d.c. neutral, 50 V line relay acting as a Track Relay (TR). A block diagram of the TI 2I receiver is shown in Figure 2.

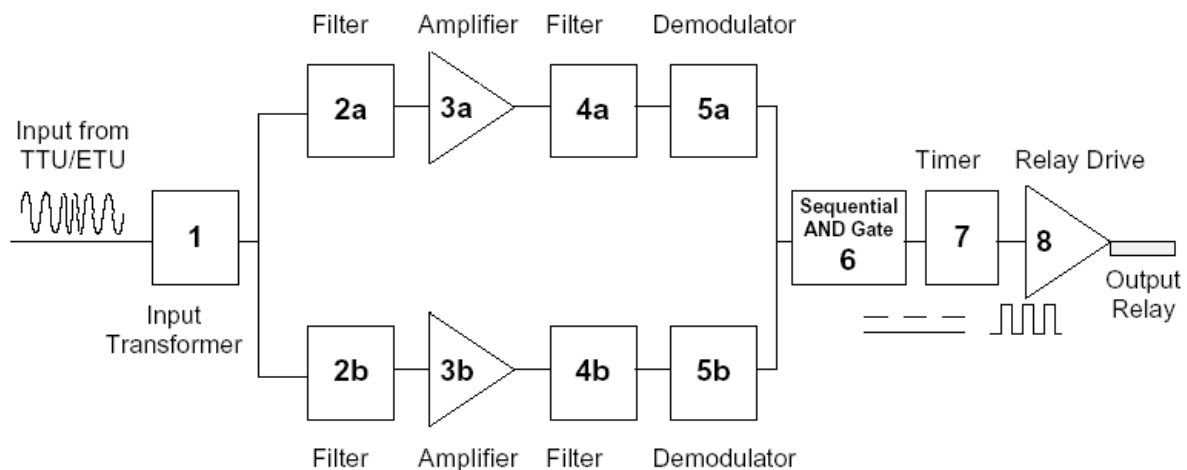


Figure 2: Block diagram of the TI 21 receiver

The input signal is fed from the TTU/ETU to the input transformer (1), which matches the receiver to the TTU/ETU and provides means of setting the overall receiver gain. The receiver has two parallel channels; one tuned to the higher frequency and one to the lower frequency. In each channel, the signal is filtered (2), amplified (3), filtered (4) and finally demodulated (5) so that a pair of anti-phase square waves are applied to the sequential AND gate (6).

The sequential AND gate only gives the required constant negative output when both the square waves are present in anti-phase to each other. This negative output is fed to a capacitor timer (7) which after a delay applies a nominal 50 V d.c. drive to an external output relay (8).

Immunity from interference is achieved because the presence of both frequencies correctly modulated is necessary for at least 2 seconds before the relay is energised [2]. Absence of the receiver input, due to a train shunt or any other reason, for a short period will disable the relay drive output.

The receiver will respond to signals outside the specified modulation range of the transmitter. In the early stages of TI 21 development the receiver was required to respond to a variety of modulation rates as part of a cab signalling/ATP system and the receiver design reflects this. The receiver will respond to modulation rates between 1.7 and 36 Hz [4]. The receiver can operate with an input signal duty cycle of up to 1:18, however this cannot be sustained at extremes of modulation rate [4].

4.2.3 TI 2I Track Tuning Unit

A TTU is used to connect a transmitter or receiver to the rails where two TI 2I track circuits meet. Together with the inductance of the rails in the tuned zone, it forms a tuned circuit to define the end of the track circuit, and contains a transformer to isolate the rail connections from the transmitter or receiver. The tuning unit is tuned to act as a short circuit to the adjacent track circuit frequency. Only track circuits of designated frequency pairs can be joined by a tuned zone, so each tuning unit only has to be tuned to short circuit a single nominal frequency. Therefore the boundary between two track circuits is defined by two TTUs connected to the rails 20 m apart. The end of a track circuit can be regarded as being mid-way between the two TTUs [2]. Figure 3 shows a typical arrangement of tuning units at the boundaries between TI 2I track circuits.

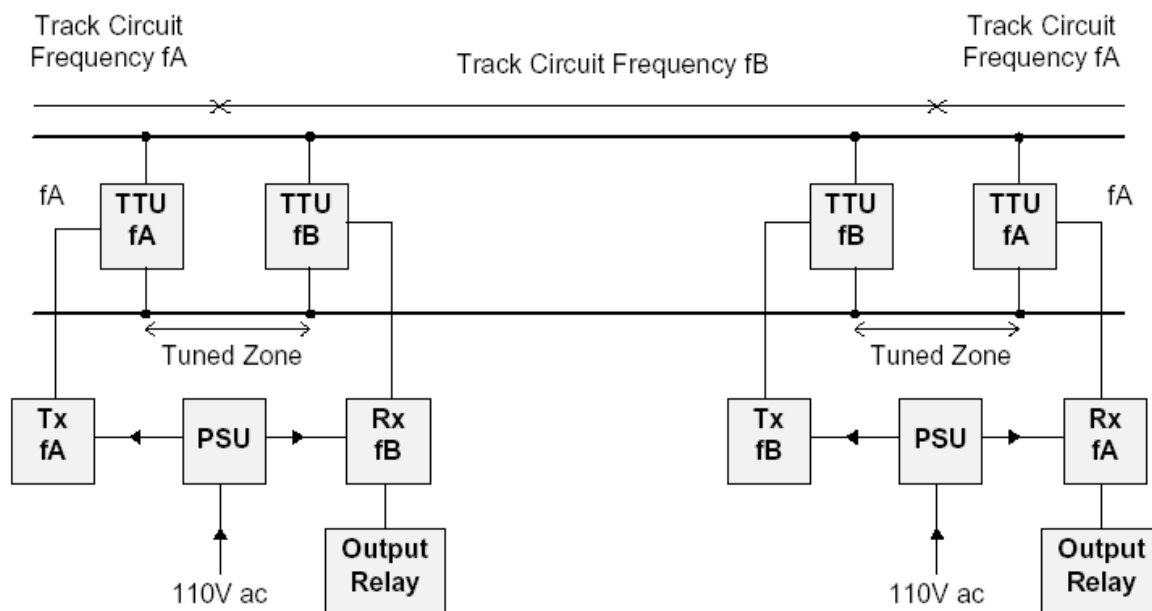


Figure 3: Basic configuration of TI 2I track circuit

4.2.4 TI 2I End Termination Unit

An ETU takes the place of a TTU and rail inductance where a TI 2I track circuit abuts a single rail track circuit or non track-circuited line. An insulated joint in each rail defines the end of the track circuit, and the traction current bonding is maintained by the use of an impedance bond. An end termination unit is also used to connect the transmitter in a centre-fed configuration. Figure 4 shows how the end termination unit is used at the boundary between a TI 2I and single rail track circuit.

4.2.5 TI 2I Impedance Bond

Traction return-current bonding is achieved by using impedance bonds. Impedance bonds resonate at the nominal frequency of each track circuit. They are also positioned where the TI 2I track circuit abuts a single rail track circuit or non track-circuited line. Each impedance bond shunts the track circuit with an impedance of about $12\ \Omega$ at TI 2I frequencies. Hence, the number, which can be connected to any track circuit, is limited.

Figure 5 shows the different ways in which the impedance bond is connected to the rails. Arrangement (a) is preferred, as the failure of any bond connection is a revealed failure, though not necessarily on the a.c. railway. However, arrangement (b) and (c) are known to exist in practice.

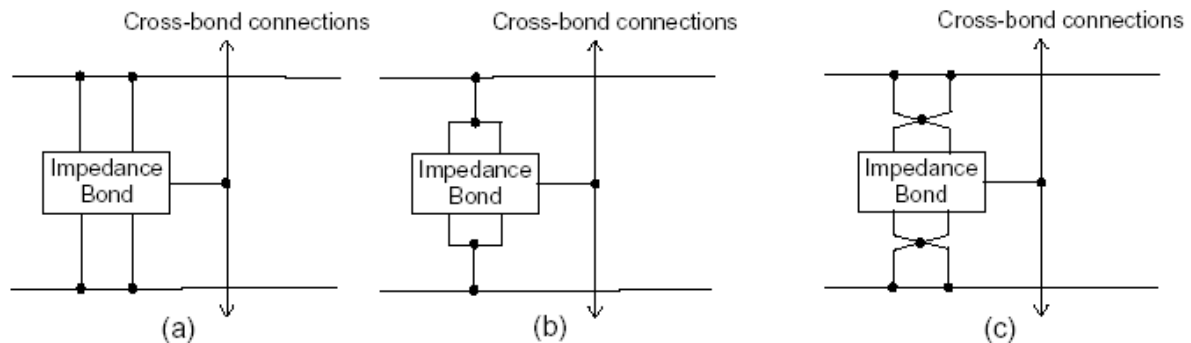


Figure 5: The arrangements of TI 2I impedance bond for cross-bond connections

Details of impedance bond equivalent circuits are shown in figures 9 and 10.

5.0 SUSCEPTIBILITY LIMITS

5.1 Transient Interference

The TI 2I track circuit requires a modulated signal to be applied to the receiver for nominally 2 seconds for it to energise. This is due to the receiver's in built time delay of at least 2 seconds before it will indicate that the track is clear [6]. Hence the track circuit is inherently immune to transient interference that does not broadly mimic the transmitter signal which persists for less than 2 seconds.

5.2 Steady state limits

The minimum rail current at TI 2I frequencies required to energise the track circuit

receiver can be calculated by modelling the track circuit receiver and its associated components. The permitted interference current level is derived from this value by applying appropriate interference allowances. Higher levels of interference current can be permitted provided modulation at rates known to energise the receiver do not persist for long enough to cause a wrong side failure of the signalling system.

The track circuit components described in section 4.2 must be represented by circuit equivalents in the model.

It is only necessary to model the track as a pair of rails - the track is modelled independently of the electrification system, since ballast resistance need not be modelled. Each rail is simply modelled as a series resistance and inductance since the frequency range of interest is relatively narrow.

It is first necessary to calculate the rail current at the beginning of the tuned zone (or at the end termination unit) to operate the receiver. Receiver input sensitivity is the only defined input level for the track circuit. Interference current is injected directly into the rails i.e. no parallel 'shunting' paths are assumed.

5.2.1 Notes on the modeling

General points to note regarding the models presented in this section are:

- Component values are contained within Appendix A
- These component value quoted are the nominal values
- Test points within the models are provided to measure train generated interference and the current flowing in the receiver
- Modelling of the TI 21 receiver is suitable due to the purely electronic nature of the sensing within the receiver and has been successfully used in the past [4]

5.3 Interference Allowances

An interference allowance is applied to the permitted interference levels to allow for additional interference sources.

For conducted interference, an interference allowance of 50% is used [4] (i.e., the allowance for additional sources of interference is $\frac{1}{2}$ of the total).

For axle-to-axle voltages, an interference allowance of 33% is used [4] (i.e., the allowance for additional sources of interference is $\frac{1}{3}$ of the total and the permissible axle to axle voltage at the train is $\frac{2}{3}$ of the total). This is discussed below.

Interference allowances take account of the following interference sources:

- Transmitter feed-through (from the same track circuit)
- Transmitter feed-through (from other track circuits of the same frequency – see below)
- Axle-to-axle voltage and conductive interference from trains on parallel tracks
- Trains remote from the track circuit but in the same feeder section
- General background noise from other sources

The level of transmitter feed-through is largely dependent on the train shunt impedance, and increases as the impedance rises. Any interference generated by the train must be at the same frequency, in phase, and modulated at the same rate as the transmitter for this to be a problem - this is most improbable.

Feed-through from other track circuits is limited to 8 mA at the receiver with infrastructure intact, and is minimised by the design and layout of the track circuit. The level of current flowing in the track circuit receiver at the operating frequency is measured when the track circuit is commissioned with the transmitter switched off. If this current exceeds 8 mA the test is deemed to have been failed and the track cannot be commissioned until it has passed this test. The level of interference current flowing in any particular track circuit is recorded on the Track Circuit Record Card [2]. Disconnection of an adjacent tuning unit will be a 'revealed failure' but may increase feed-through until repaired. However, the probability of the train producing interference that matches the feed-through in frequency, phase and modulation rate is assumed to be negligible, especially when the duration of this required to cause a wrong-side failure (2 seconds) is taken into account.

Axle-to-axle voltage from a train on a parallel track can only have a significant effect whilst the train is aligned with cross bonds.

The contribution from trains remote from the track circuit will be small. These sources must be at the same frequency, in phase, and modulated at the same rate as interference from the train being analysed for this to be a problem - this is most improbable.

General background noise from other sources may be considered insignificant, because of the requirement to match the frequency, phase, and modulation rate of the interference generated by the train.

Variations in values of critical impedance bond and tuning unit components including length of cables and tuned zones are accounted for in the sensitivity analysis. The results are given below.

5.4 Current Threshold for TI2I track circuits

The level sensing in the TI 2I receiver is purely electronic. Pick-up and drop-away current levels are the same. This approximates to no hysteresis within the receiver. The receiver gain setting is dependent on track circuit length and the presence of impedance bonds. The maximum gain setting is step 13, which corresponds to an energisation current of 15 mA at the receiver input [4]. For drop-away to be significant, the track circuit must first be energised by a transmitter and subsequently held up by interference (i.e., by the interference exceeding the energisation level without a break, the frequency and modulation both being in phase). This is considered improbable. For the purposes of this analysis, centre-fed track circuits can be treated as a pair of end-fed track circuits. If a signal below the 15mA threshold is applied to the track circuit receiver, it will not energise the receiver regardless of any modulation.

5.5 Train current limits

Train current limits will be calculated for the worst case imbalance of 100% - effectively a failed infrastructure condition.

5.5.1 Track circuit with ETU

A typical track circuit model, produced in order to specify the maximum allowable conductive interference levels in a TI 2I track circuit with ETU is shown in figure 6. The circuit shown is an ETU for frequency code A and receiver. Since the assumption of 100% imbalance requires an impedance bond side lead disconnection, it is not necessary to model the impedance bond.

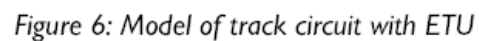


Figure 7 shows a circuit diagram of a model to analyse a track circuit with TTU. The circuit shown is a TTU for frequency code A and receiver and a TTU for frequency code B and receiver, separated by a length of track.

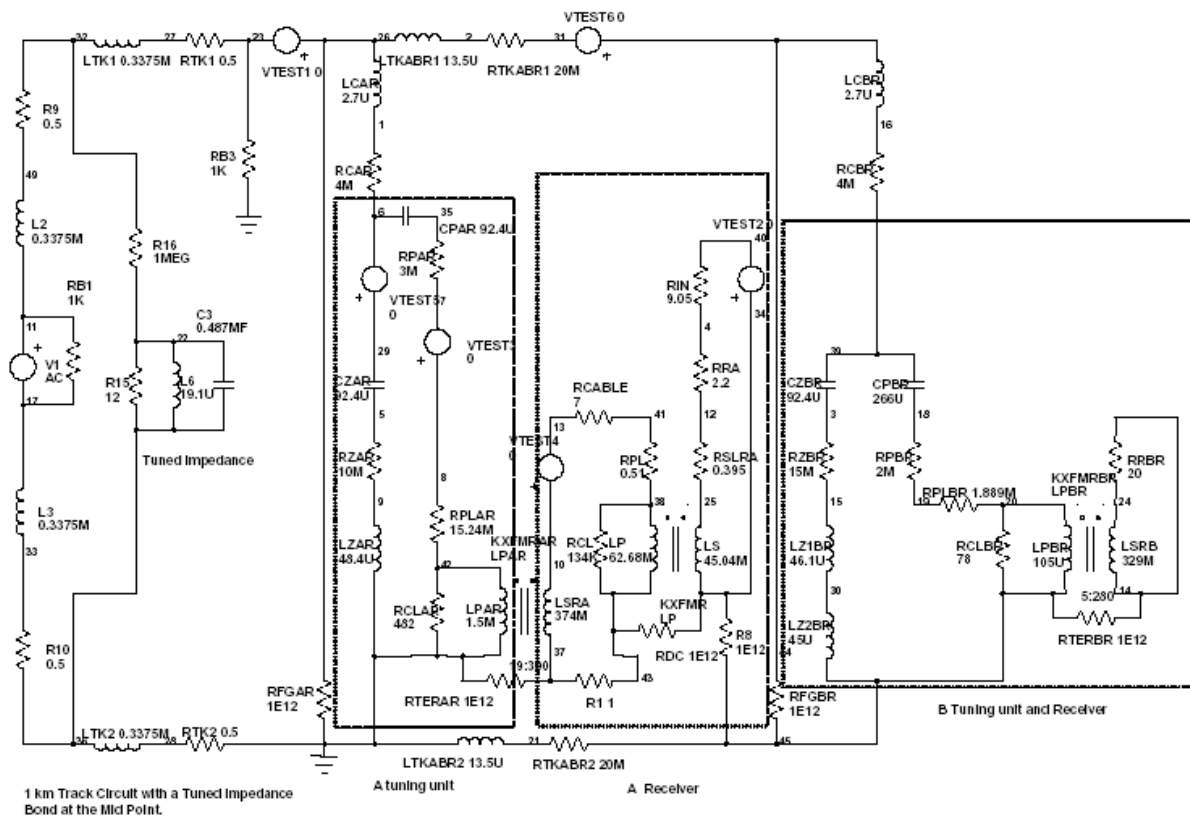


Figure 7: Model of track circuit with TTU

5.5.3 Calculation of train current limits

A series of models were used, of which figures 6 and 7 are examples used to analyse all possible track circuit configurations with TTU/ETU changed to suit different operating frequencies.

The methodology to determine the permissible rail current is as follows.

- For the models shown in figures 6 and 7, the voltage $V1$ was set to 5.5 V a.c..
- Rail current is measured at $VTEST1$ and the receiver input current at $VTEST4$.
- These currents are calculated at the lower and upper frequencies for each track circuit frequency code.
- At each frequency the rail current needed to produce a receiver input current of 15 mA is calculated using the following equation:

$$\text{Rail Current (mA)} = \frac{I(VTEST1)}{I(VTEST4)} \times 15 \quad (1)$$

The circuit models shown in figures 6 and 7 are run for each of the TI 2I frequencies. The model shown in figure 7 was also run for each of the permitted frequencies and configurations of the track circuit receiver and adjacent TI 2I track circuit. For each run, the component values specified in Appendix A are used.

In addition, the effects of inter-rail capacitance and a rail break should be investigated. Interrail capacitance is modelled by adding a capacitor to the model. In the present example a figure of 320 nF is inserted between the rails in the tuned zone (this is equivalent to 16 μ F/km). A rail break is modelled by placing a 10 M Ω resistor in series with one of the resistors representing a rail in the tuned zone.

5.5.4 Train current limits for track circuit with ETU

Table 3 shows the rail current which is required to produce an input current of 15 mA at the receiver for a typical scenario. These results were obtained using the nominal values of track circuit parameters. Since both frequencies must be present with the correct modulation for the receiver to energise, it is only necessary to ensure that the input at one of the frequencies is not present to prevent the receiver energising. Thus the minimum current required to energise the receiver is the higher current required at either of the pair of frequencies. It can be seen from the table that in each case, the upper frequency requires a higher current, and this is the permissible limit. The use of the higher permissible current limit is justified due to the pessimistic assumptions made in the modelling process listed in section 6.

Table 3 shows that a track circuit with ETU for frequency code C needs the least interference current to energise the receiver.

Frequency code of ETU being analysed	Rail Current, mA, at frequency code of track circuit being analysed (lowest current in bold)	
	Lower Frequency	Upper Frequency
A	178.9	202.2
B	237.5	264.0
C	157.6	182.8
D	314.0	334.3
E	242.6	268.7
F	269.1	295.6
G	231.6	284.2
H	261.4	285.8

Table 3: Rail current required for a receiver input current of 15 mA
(using nominal values of track circuit with ETU)

5.5.5 Train current limits for track circuit with TTU

Table 4 shows the rail current required to energise the receiver. Using nominal values of the track circuit parameters, the results show that the configuration of a TTU for frequency code C abutting a TTU for frequency code D needs the least current to energise the receiver.

Track Circuit Frequency Code		Rail Current, mA, at frequency of track circuit being analysed (lowest current in bold)	
Rx being analysed	Rx (adjacent tuning unit)	Lower Frequency	Upper Frequency
A	B	205.9	233.0
C	D	176.9	209.2
E	F	226.6	249.0
G	H	189.3	219.6
B	A	246.9	268.5
D	C	230.0	264.8
F	E	228.8	262.8
H	G	225.1	247.4

Table 4: Rail current required for a receiver input current of 15 mA
(Using nominal values of track circuit with TTU)

5.5.6 Right Side Failure Condition

From ref. 7, the minimum transmitter signal current in the rails is 3.4 amps. It follows therefore that the interference current would have to reduce this below the threshold level. To do this the interference would have to be at the specific channel operating frequency and be in anti-phase to it. Under nominally balanced conditions the level of current is in excess of that required to cause a wrong side failure (182.8 mA) under the 100% imbalance condition as calculated in section 5.5.5. Since this permissible level is based on a pure tone rather than the modulated signal required to 'pick up' a TI 2I track circuit receiver this level can be compared to that required to cause wrong side failure. This permissible current represents only 5.4% of the minimum transmitter signal in the rails the right side failure case is encompassed in this level.

5.5.7 Worst case train current limits

To specify the maximum permitted interference current for TI 2I track circuit, the worst case must be considered. From Tables 3 and 4 it can be seen that a track circuit represents this case with ETU for frequency code C.

Applying the 50% interference allowance, the maximum rail current permitted by a single train becomes 91.4 mA. This represents a 50% of the 182.8 mA calculated for the upper frequency of a track circuit with ETU for frequency code C.

This is calculated using the nominal component values given in Appendix A and the pessimistic assumptions listed in section 3. The interference allowance will also encompass variation due to component drift.

5.6 Calculation of permitted axle-to-axle voltage

In a TI 2I track circuit installation, all cross bonds are connected via resonated impedance bonds, as described above. Unless multiple failures are present, there is little possibility of TI 2I frequencies circulating through other tracks. For the axle-to-axle voltage to propagate to a receiver it would require significant imbalance in the track circuit. The worst case scenario has been identified as a current circulating via a pair of adjacent cross bonds, and the receiver being falsely energised by the interference on a track circuit which is 100% unbalanced.

5.6.1 Axle - Axle Voltage Model

One scenario to be investigated is a rail break within the track circuit and a train on the adjacent track producing an axle-to-axle voltage. This would energise the track circuit receiver, causing the rail break to be undetected and therefore causing any train entering the track circuit to remain undetected.

Figure 8 shows the worst case scenario as described above. A minimum cross bond spacing of 100 m [4] is used in this model. This means that a minimum rail impedance is considered, hence the model is more sensitive to axle-to-axle voltage. The distance between the axles of the train producing the axle-to-axle voltage is assumed the same as the distance between the cross bonds. The tracks are terminated by two sets of impedance bonds.

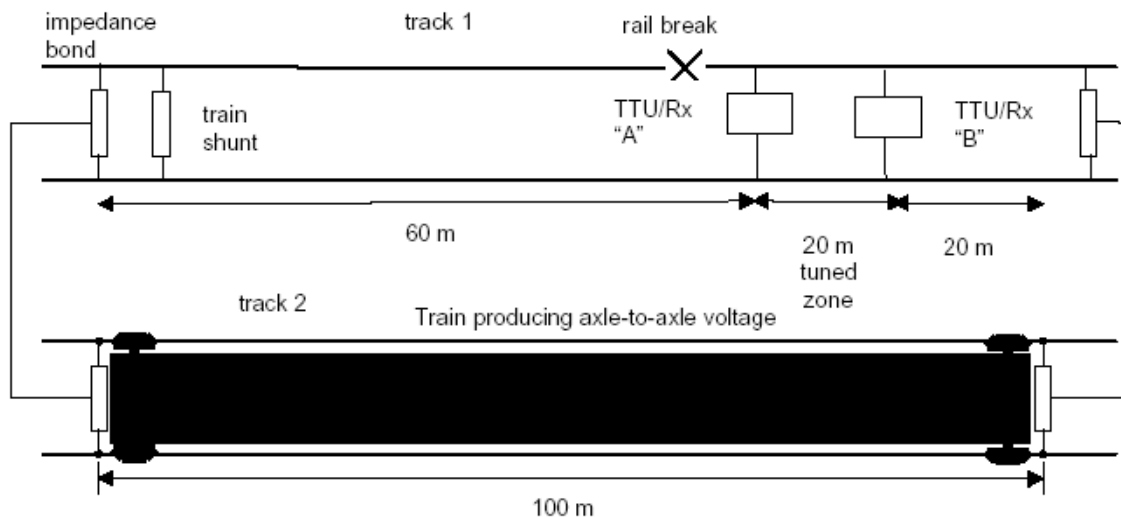


Figure 8: Basic configuration of the case considered for axle-to-axle voltage analysis

Figures 9 and 10 show the circuit models that correspond to the two different types of impedance bonds applying to a.c. and d.c. electrified lines. The model includes the tuned zone consisting of A and B type TTUs and receivers at one end of the section on track 1 (see point d below).

The train on the track circuit affected by interference is represented by a 0.1Ω train shunt at the impedance bond furthest from the break. This is regarded as a reasonable worst case.

Further assumptions are:

- (a) The interfering train is a pure voltage source of zero impedance. This is extremely pessimistic and assumes that the train has zero impedance.
- (b) The TI 21 receiver is operating at the maximum sensitivity, i.e. 15mA is the threshold current.
- (c) The interfering train is stationary, and exactly fills the space between the two impedance bonds. This again is very pessimistic. Since the TI 21 receiver requires a

correctly modulated input for at least 2 seconds, the probability of this happening with a stationary train is very low.

(d) The modelling is carried out at all frequencies, but the worst case is code A for the track circuit containing the break. This is the worst case [4].

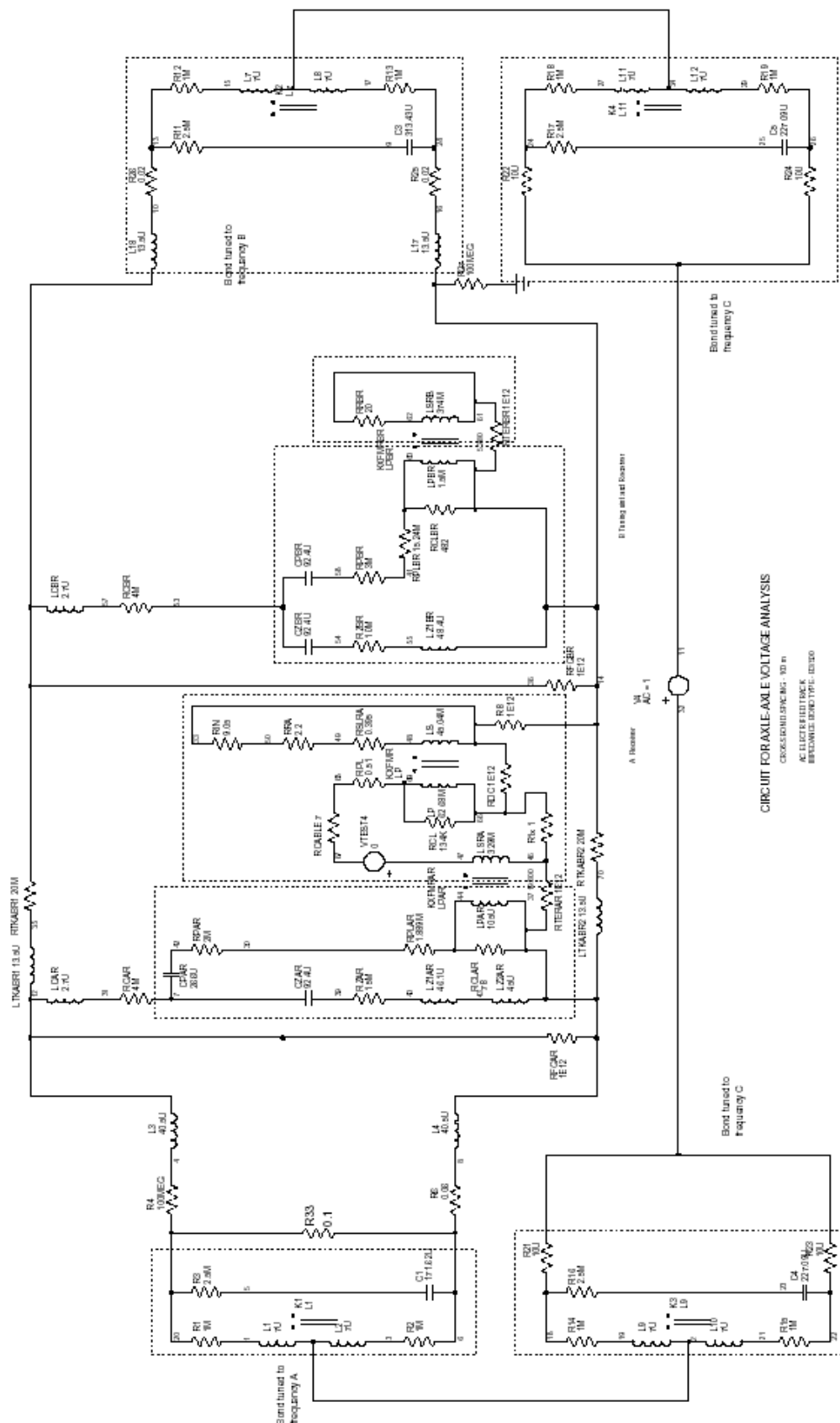


Figure 9: Model used to calculate the permitted axle-to-axle voltage
(using B3/500 impedance bond)

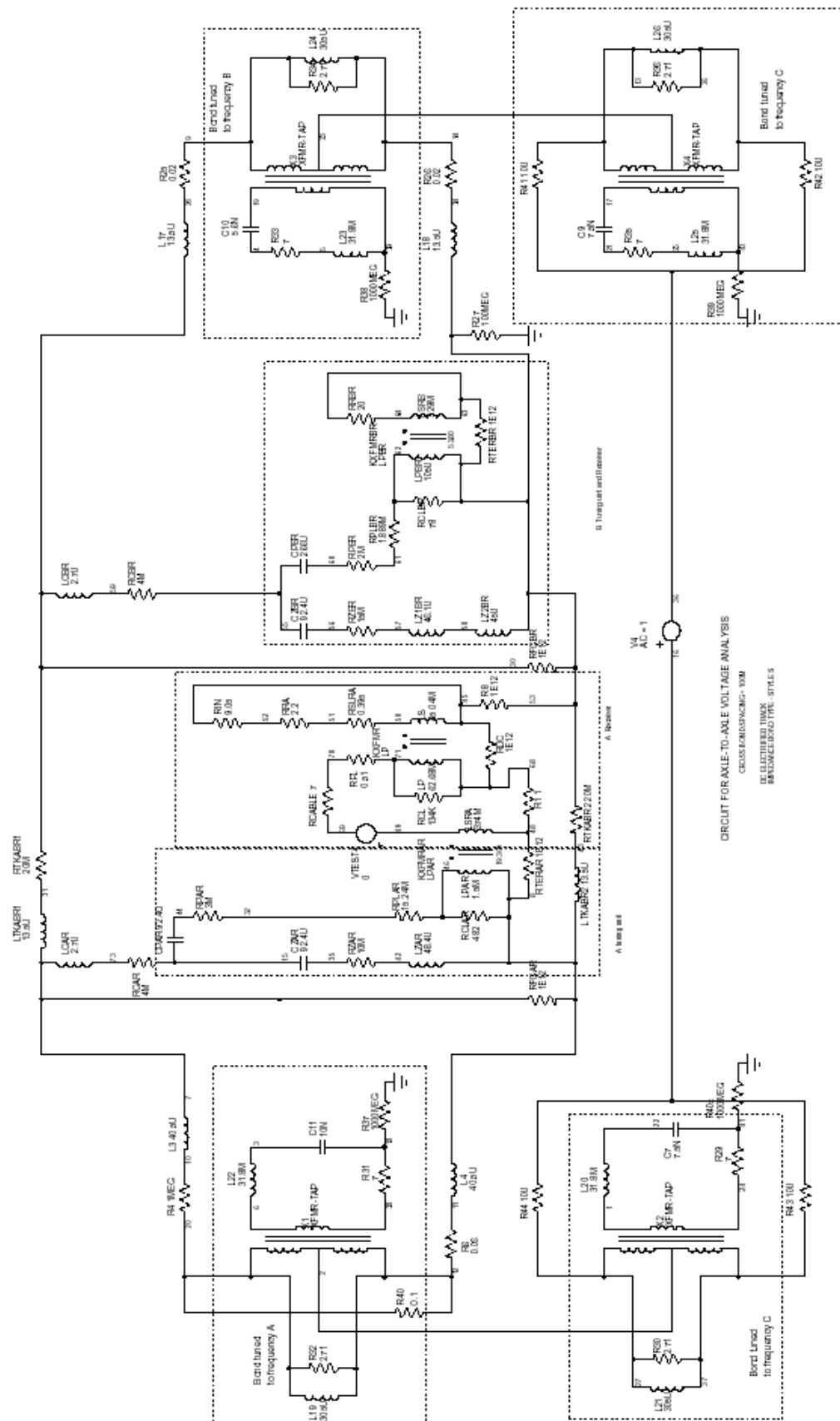


Figure 10: Model used to calculate the permitted axle-to-axle voltage
(using Style S impedance bond)

5.6.2 Permissible axle - axle voltage calculation

The track circuit models shown in figures 9 and 10 can be modelled using a circuit analysis program. The methodology to determine the maximum permissible axle - axle voltage is as follows:

- The voltage V_4 was set to 1 V a.c. and the models is run using all permitted frequency code combinations.
- The rail current at V_{TEST4} was then measured at two frequencies, the lower and upper frequencies for the frequency code of the track circuit containing the broken rail.
- At each frequency the interference voltage needed to produce the maximum receiver input current of 15 mA is calculated using the following equation:

$$\text{Axle-to-axle voltage (mV)} = \frac{1}{I(V_{TEST4})} \times 15 \quad (2)$$

The impedance bonds on each track are tuned to the frequency code of the corresponding track.

5.6.3 Axle to Axle Voltage Limits

The axle-to-axle voltage analyses are carried out using the frequency codes A-D which are the preferred values for areas with two tracks only. Areas with more than two tracks are less sensitive to interference as they have inherently more parallel return paths for interference currents.

Tables 5 and 6 below show the minimum axle-to-axle voltage required to produce a receiver input current of 15 mA when d.c. or a.c. type impedance bonds were used respectively. These results were obtained using nominal values of the impedance bond parameters.

Frequency code of the broken rail	Frequency code of the adjacent track	Axle-to-axle voltage, mV, at frequency code of the broken rail (limiting voltage in bold)	
		Lower Frequency	Upper Frequency
A	C	225.2	258.9
A	D	225.2	258.9
B	C	342.3	372.6
B	D	342.3	372.6
C	A	219.7	261.1
C	B	219.7	261.1
D	A	347.9	410.2
D	B	347.9	410.2

Table 5: Axle-to-axle voltage required for a receiver input current of 15 mA
(Using nominal values of the Style S impedance bond parameters)

Frequency code of the broken rail	Frequency code of the adjacent track	Axle-to-axle voltage, V, at frequency code of the broken rail (limiting voltage in bold)	
		Lower Frequency	Upper Frequency
A	C	226.2	259.1
A	D	226.2	259.1
B	C	316.3	364.2
B	D	316.3	364.2
C	A	220.5	261.6
C	B	220.5	261.6
D	A	344.3	408.3
D	B	344.3	408.3

Table 6: Axle-to-axle voltage required for a receiver input current of 15 mA (Using nominal values of the B3/500 impedance bond parameters)

The two tables show that the minimum permissible axle-to-axle voltage was obtained when a broken rail was on a track circuit with frequency code A. The frequency code of the track circuit on the adjacent track has no effect on the results.]

As with the conductive interference, the minimum axle-to-axle voltage is the higher voltage required at either frequencies. From Tables 5 and 6 it is clear that the maximum permitted axle-to-axle voltage at TI 21 frequencies between any pair of axles, V_{AA} is 259 mV when the d.c. impedance bond is used.

This result represents a maximum permissible axle-to-axle voltage of 173 mV, allowing an interference allowance of 33% for other sources of interference.

The maximum permitted axle-to-axle current I_{AA} can be calculated from V_{AA} as:

$$I_{AA} = \frac{V_{AA}}{Z_{TR}} \text{ mA} \quad (3)$$

i.e.

$$I_{AA} = \frac{173}{Z_{TR}} \text{ mA using a 33% interference allowance.} \quad (4)$$

Where Z_{TR} represents the appropriate train impedance between the pair of axles for which the calculation is made. The result will be pessimistic, as it does not include the additional load of the circuit external to the train.

6.0 METHODOLOGY FOR DEMONSTRATION OF COMPATIBILITY

It must be confirmed that the track circuits are installed in double rail, jointless mode only.

The analysis is based on the most sensitive receiver setting. There is therefore no requirement to place any restrictions on future adjustment of track circuit receivers on the route.

6.1 Normal train

The steady state train emission data should be compared to the limits defined in section 5.5 for all normal operating conditions. These limits are based on effectively failed infrastructure which is the only situation that can give rise to the 100% imbalance situation as modelled and as such will envelope all infrastructure conditions likely to be encountered on the railway. These limits should not be breached under normal operating conditions.

If the absolute levels defined in section 5.5 cannot be met then the absence of modulation of the correct type can be used to demonstrate compatibility. It is anticipated that most traction systems will comply with the simple level comparison.

Transient interference is not an issue for TI 2I track circuits if existing for less than two seconds as discussed in section 5.1.

6.2 Faulty Train

A failure rate analysis is performed. This determines the frequency with which trains exceed the limits calculated in section 5.5. The frequency with which limits are exceeded together with the magnitude and duration of such exceedances when these exceedances are longer than the 2 second time delay and modulation requirements of the track circuit should be taken into account when constructing mitigating arguments.

6.3 Axle - Axle voltage limits

The train shall be shown to comply with the limit calculated in section 5.6 under all normal operating conditions. The exceedance rate should be calculated for all credible fault conditions.

6.4 Safety Targets

The overall exceedance rate shall be found by comparison of the emissions data with the defined limits as described above. This total exceedance rate shall be shown to be as low as reasonably practicable (ALARP) in the context of the safety requirements of the project.

APPENDIX A: Parameters

Track			
Component Label(s)	Value	Source of Value	Component Represents
V1	5.5 V	Arbitrary	Voltage source
L2, L3, LTK1, LTK2	337.5 μ H	Provided by the manufacturer	Inductance of 500 m of rail
R9, R10, RTK1, RTK2	0.5 Ω		Resistance of 500 m of rail
RB1, RB3	1 k Ω	This is an arbitrary value, as it does not affect the results of this study. It also provides a ground connection needed by the package.	Ballast resistance

Impedance Bond			
Component Label(s)	Value	Source of Value	Component Represents
L6	19.1 μ H	Calculated for impedance of 6 m Ω at 50 Hz	Impedance bond (it does not affect the results of this study)
R15	12 Ω	Provided by the manufacturer (minimum resonated impedance)	
C3	0.487 μ F	Calculated to create a tuned circuit with L6.	

Not Part of Model			
Component Label(s)	Value	Source of Value	Component Represents
RFGAR, RFGBR, RTERAR, R8, RTERBR, RDC	$1e12\ \Omega$	Required by the circuit analysis tool. It effectively creates a high impedance D.C. path to earth.	

TTU			
Component Label(s)	Value	Source of Value	Component Represents
LCAR	$2.7\ \mu\text{H}$	Provided by the manufacturer	Connecting Cables (representing both connections).
RCAR	$4\ \text{m}\Omega$		Impedance of internal components
RPAR	$3\ \text{m}\Omega$		
RZAR	$10\ \text{m}\Omega$		
RPLAR	$15.24\ \text{m}\Omega$		
RCLAR	$482\ \Omega$		
KXFMRRAR	0.9964		
LSRA	$374\ \text{mH}$	Calculated from data provided by the manufacturer	

ETU			
Component Label(s)	Value	Source of Value	Component Represents
LCAR	2.7 μ H	Provided by the manufacturer	Connecting Cables (representing both connections).
RCAR	4 m Ω		
RPAR	3 m Ω		Impedance of internal components
RZAR	10 m Ω		
RPLAR	15.24 m Ω		
RCLAR	482 Ω		
C2AR	92.4 μ F	Provided by the manufacturer	Internal circuit of ETU
CZAR	92.4 μ F		
LZAR	49.1 μ H		
L2AR	62.3 μ H		
C3AR	266 μ F		
RIAR	0.27 Ω		
R2AR	0.1 Ω		
R3AR	0.1 Ω		
LPAR	1.5 mH	Provided by the manufacturer	Output transformer
KXFMRAR	0.9964		
LSRA	374 mH	Calculated from data provided by the manufacturer	Output transformer

Receiver			
Component Label(s)	Value	Source of Value	Component Represents
RCABLE	7 Ω	Provided by the manufacturer	Resistance of receiver cable
RI	1 Ω		Series resistance for measuring receiver input current
RPL	0.51 Ω		Input Transformer
RCL	134 k Ω		
LP	62.68 mH		
LS	45.04 mH		
KXFMR	0.981		
RSLRA	0.395 Ω		
RRA	2.2 Ω	Reference 4	Impedance of receiver
RIN	9.05 Ω	Calculated from data provided by the manufacturer	

Tuned Track Section			
Component Label(s)	Value	Source of Value	Component Represents
LTKABR1, LTKABR2	13.5 μ H	Provided by the manufacturer	Inductance of tuned section
RTKABR1, RTKABR2	20 m Ω		Resistance of tuned section

TTU			
Component Label(s)	Value	Source of Value	Component Represents
LCBR	2.7 μ H	Provided by the manufacturer	Connecting Cables
RCBR	4 m Ω		
RPBR	2 m Ω		Impedance of internal components
RZBR	15 m Ω		
RPLBR	1.889 m Ω		
RCLBR	78 Ω		
KXFMRBR	0.9964		
LSRB	329 mH	Calculated from data provided by the manufacturer	
RRBR	20 Ω	Provided by the manufacturer	Input impedance of receiver

Table A.1: Parameters for Train Current Limit Model
(Reproduced from reference 4)

TTU Frequency code	CZ _n R	CP _n R	LZ _n R	LP _n R
A	92.4 μ F	92.4 μ F	48.4 μ H	1.5 mH
B	92.4 μ F	266 μ F	46.1 μ H + 45 μ H	105 μ H
C	62.5 μ F	62.5 μ F	56.3 μ H	1.5 mH
D	62.5 μ F	212 μ F	52.0 μ H + 45 μ H	105 μ H
E	117 μ F	117 μ F	43.7 μ H	1.5 mH
F	117 μ F	300 μ F	41.7 μ H + 45 μ H	105 μ H
G	75 μ F	75 μ F	52.8 μ H	1.5 mH
H	75 μ F	230 μ F	49.6 μ H + 45 μ H	105 μ H

Table A.2: Code dependent TTU Parameters

NB: The '*n*' in the parameter name stands for the relevant frequency code (as used in figures 6, 7, 9 and 10)

ETU Frequency Code	C1 _n R and C2 _n R	C3 _n R	L1 _n R	L2 _n R	LP _n R
A	92.4 μ F	266 μ F	49.1 μ H	62.3 μ H	1.5 mH
C	62.5 μ F	212 μ F	56.3 μ H	59.5 μ H	1.5 mH
E	117 μ F	300 μ F	59.0 μ H	64.8 μ H	1.5 mH
G	75 μ F	230 μ F	51.9 μ H	76.4 μ H	1.5 mH

ETU Frequency Code	C1 _n R	C2 _n R and C3 _n R	L1 _n R	L2 _n R	LP _n R
B	Not present	266 μ F	Not present	34.3 μ H	105 μ H
D	Not present	212 μ F	Not present	38.2 μ H	105 μ H
F	Not present	300 μ F	Not present	36.7 μ H	105 μ H
H	Not present	230 μ F	Not present	36.8 μ H	105 μ H

Table A.3: Code dependent ETU Parameters

NB: The '*n*' in the parameter name stands for the relevant frequency code (as used in figures 6, 7, 9 and 10)

Westinghouse Style S Impedance Bond			
Component Label(s)	Value	Source of Value	Component Represents
X1, X2, X3, X4		Ideal transformer	Transformers
C7, C9	7.5 nF	Reference 4	Capacitors
C10	5.6 nF		Capacitor
C11	10 nF		Capacitor
L20, L22, L23, L25	31.8 mH	Reference 4	Inductors
R30, R32, R34, R36	2.71 Ω		Resistors
R29, R31, R33, R35	7 Ω		Resistors
L19, L21, L24, L26	305 μ H	Calculated value (impedance bonds are tuned to the corresponding frequency code)	Inductors

Train			
Component Label(s)	Value	Source of Value	Component Represents
V4	1 V a.c.	Arbitrary.	Interference produced by train.

Track Circuit Receivers and TTUs
The component values and derivations for these are the same as those given in Table A.1, A.2 and A.3.

Table A.4: Parameters for Axle - Axle Voltage Model
(Reproduced from reference 4)

APPENDIX B: Assumptions

General

The TI 2I track circuit requires a modulated signal to be applied to the receiver for nominally 2 seconds for it to energise. This is due to the receiver's in built time delay of at least 2 seconds before it will indicate that the track is clear [6]. Hence the track circuit is inherently immune to transient interference that does not broadly mimic the transmitter signal, which persists for less than 2 seconds.

A broken signal rail will lead to a protective RSF.

Configuration

The track circuits are assumed to be installed in double rail mode.

The track circuits are assumed to be installed in jointless mode.

The effect of other parallel paths for return current that may exist are ignored. This is a pessimistic assumption.

All of the return current flows in the rails - there is no leakage between the rails. This is a pessimistic assumption.

Inter-rail capacitance is modelled as a 320 nF capacitor inserted between the rails in the tuned zone

Infrastructure failure modes

The infrastructure condition assumed (100% imbalance) is effectively the worst case fault condition for a TI 2I track circuit. It has been assumed that only one infrastructure failure can occur on one track circuit at one time.

APPENDIX C: Hazards

Interference current in traction return current

This hazard is covered by the assessment as presented.

Axle to Axle or Longitudinal voltage

This hazard is covered by the assessment as presented.

Direct induction into track mounted or trackside components

Alternating magnetic fields at TI 21 frequencies need to be considered for possible induction effects.

Particular transient events

The TI 21 track circuit requires a modulated signal to be applied to the receiver for nominally 2 seconds for it to energise. This is due to the receiver's in built time delay of at least 2 seconds before it will indicate that the track is clear [6]. Hence the track circuit is inherently immune to transient interference that does not broadly mimic the transmitter signal, which persists for less than 2 seconds.

Wheelset impedance

The maximum wheelset impedance is not quoted in this document

Vehicle geometry (overhang, wheelbase etc..)

This hazard is not covered by this document.

RSF due to excessive off channel current

This hazard is not covered by this document.

APPENDIX D: Mitigating Factors

Pessimistic assumptions

Zero train length for line current limits.

Train length equal to cross bond spacing for longitudinal voltage limit.

Rail return only for traction current.

All of the return current flows in the rails.

A worst case train position is considered in the analysis

Assumptions/variations covered by interference allowance

Normal variation of track circuit receiver performance.

Background/coincident sources of interference covered by interference Allowance

Other rolling stock.

Transmitter breakthrough.

APPENDIX E: Sources of Technical Work

The technical work contained in this document is taken from work carried out for the ALSTOM Juniper project by AEA Technology Rail, the ISA was WS Atkins Rail.

Some technical details of the track circuit is taken from the Class 92 EMC safety case, produced by AEA Technology Rail, ISAs were WS Atkins Rail (secretariat reviewer) and Mott MacDonald.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019

Issue 1

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
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